Student Performance Analysis

Author: Mariam
Role: Data Analyst (in training)
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Project Description

This project analyzes student performance data to identify factors that influence academic success. The analysis focuses on gender and parental education level.

Tools Used

- Python
- Pandas
- Matplotlib

In [7]:
```
import pandas as pd
import numpy as np

# إنشاء بيانات طلاب بشكل مباشر
np.random.seed(42)

data = {
    "gender": np.random.choice(["male", "female"], 100),
    "parental level of education": np.random.choice([
        "high school", "bachelor", "master"
    ], 100),
    "math score": np.random.randint(50, 100, 100),
    "reading score": np.random.randint(50, 100, 100),
    "writing score": np.random.randint(50, 100, 100)
}

df = pd.DataFrame(data)

df.head()
```

Out[7]:

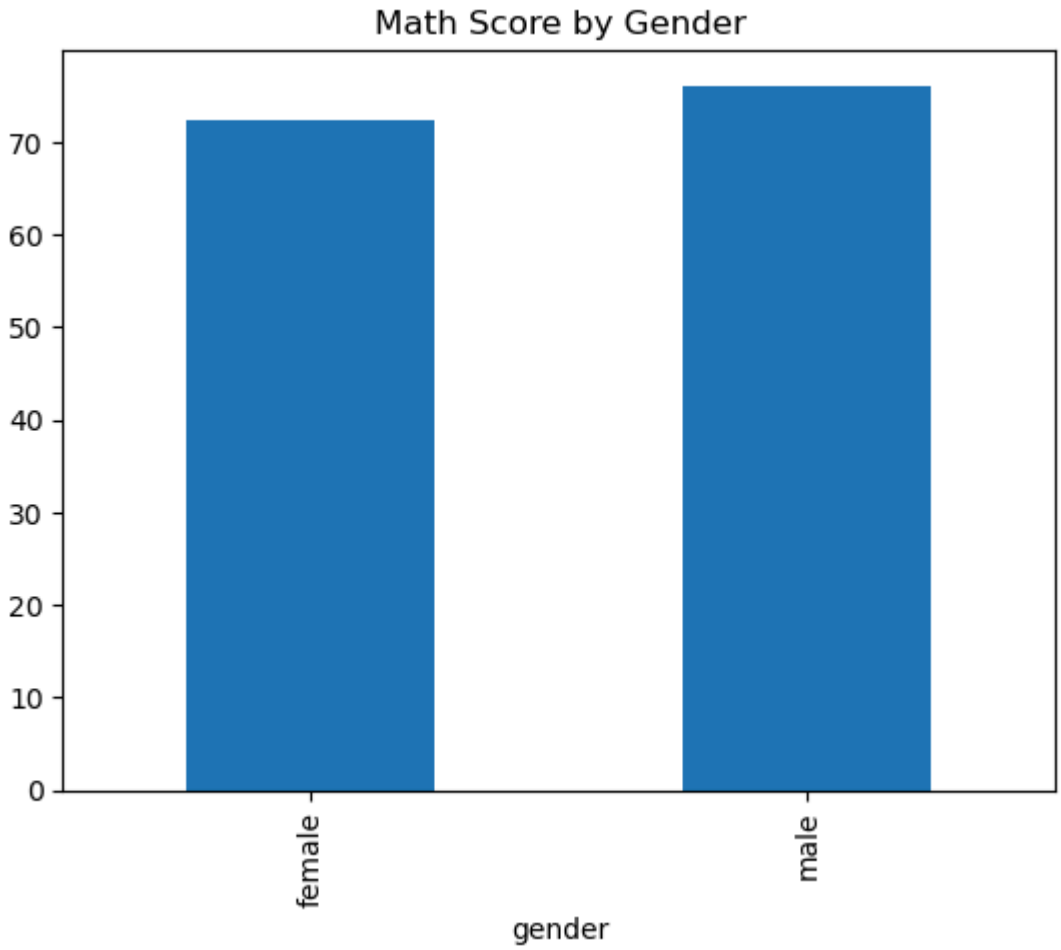
	gender	parental level of education	math score	reading score	writing score
0	male	master	77	56	72
1	female	bachelor	77	65	82
2	male	bachelor	93	75	52
3	male	bachelor	93	97	67
4	male	bachelor	69	98	74

In [11]:
```
df.groupby("gender").mean(numeric_only=True)
```

Out[11]:

gender	math score	reading score	writing score
female	72.410714	76.946429	77.285714
male	76.090909	76.363636	74.045455

In [16]:
```
df.groupby("gender")["math score"].mean().plot(kind="bar")
plt.title("Math Score by Gender")
plt.show()
```



Insight

There is a slight difference in math performance between male and female students, indicating that gender may have a minor influence on academic outcomes.

In [12]:
```
df.groupby("parental level of education").mean(numeric_only=True)
```

Out[12]:

parental level of education	math score	reading score	writing score
bachelor	76.500000	74.269231	75.923077
high school	70.742857	76.285714	78.942857
master	75.333333	78.666667	73.051282

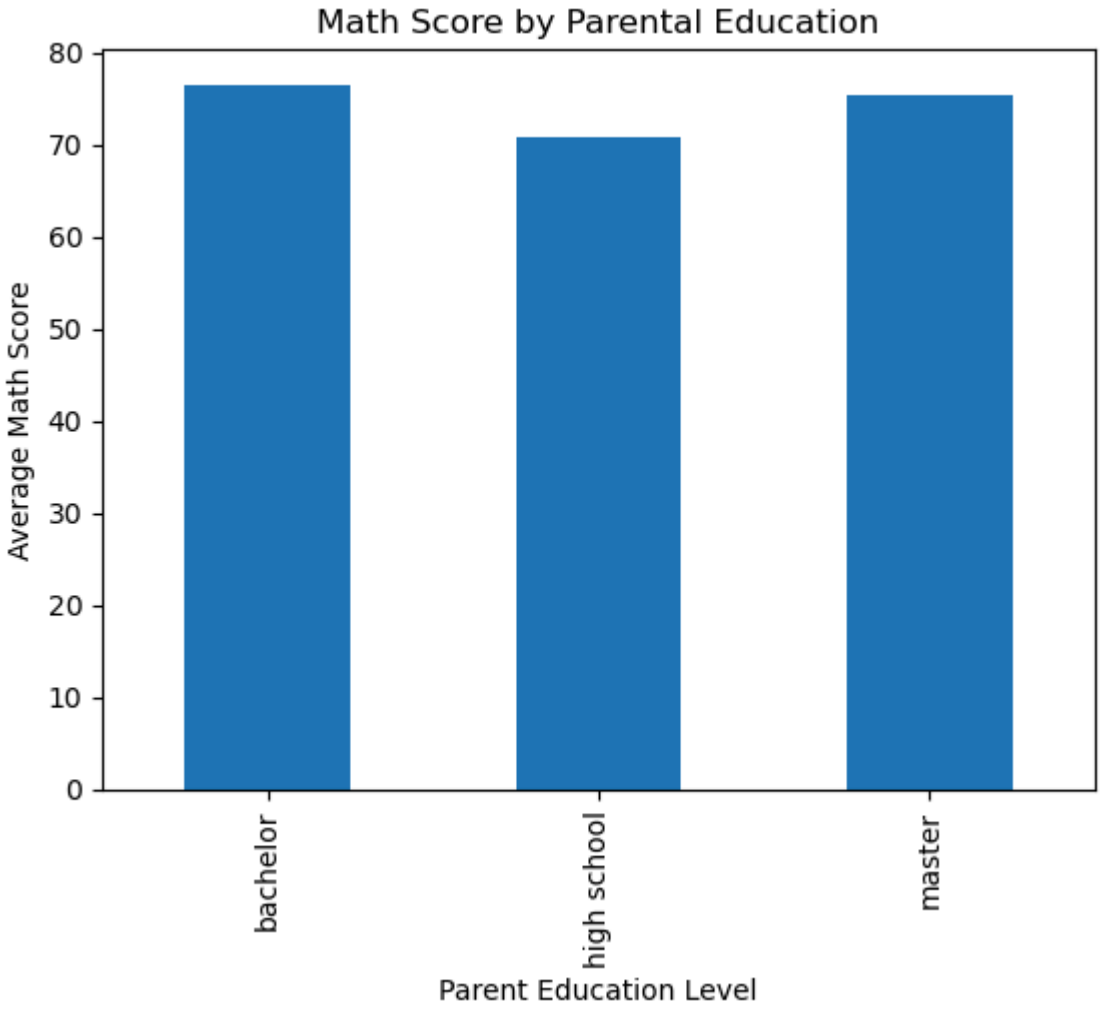
In [13]:

```
import matplotlib.pyplot as plt

df.groupby("parental level of education")["math score"].mean().plot(kind="bar")

plt.title("Math Score by Parental Education")
plt.xlabel("Parent Education Level")
plt.ylabel("Average Math Score")

plt.show()
```



Key Insight

The analysis shows that students with more highly educated parents tend to perform better academically, especially in math. While gender differences exist, they are relatively small, suggesting that parental education may have a stronger influence on student performance.

Conclusion

The analysis revealed that parental education level plays a role in student performance, especially in math. While gender differences exist, they are relatively small. These insights can help educators focus on strategies that support students more effectively.

Prepared by: Mariam
Field: Data Analysis